

Lab 6 – Configuring Network Connectivity

Lab submission

Your lab submission should contain the following three files:

File 1:

1. Create a document called **Lab6_firstname_lastname**, where *firstname* is your first name and *lastname* is your last name.
2. Put your full name at the top of the document.
3. Take screen captures (Use the Windows Snipping Tool) as requested in red text with the tag **<<SS##>>**, label each screen capture with the entire tag and description in red.
4. If any questions are asked, they will be labelled as **<<Q##>>** in red text as well, insert the question tag and answer into your document in order.
5. When finished submit the document, which only contains the labelled screenshots, as a **.pdf** to **Assessments>Dropbox>Lab 6** prior to the posted due date.

File 2:

1. Make sure each screen shot is saved as SS##, where the octothorps are the appropriate number.
2. Zip all the pictures into one file and submit to the Brightspace Dropbox.
3. Note: These pictures contain metadata which can be used for verification if any questions arise regarding academic misconduct.

File 3:

1. A **.pdf** copy of the lab document with the tables completed for Problems 1-5 in Part A.

A. Calculate IPv4 Subnets

Fill out the tables below with appropriate answers given the IPv4 address, original subnet mask, and new subnet mask.

Problem 1:

Given:	
Host IP Address:	192.168.200.139
Original Subnet Mask	255.255.255.0
New Subnet Mask:	255.255.255.224

Find:	
Number of Subnet Bits	3
Number of Subnets Created	8
Number of Host Bits per Subnet	5
Number of Hosts per Subnet	32
Network Address of this Subnet	192.168.200.128
IPv4 Address of First Host on this Subnet	192.168.200.129
IPv4 Address of Last Host on this Subnet	192.168.200.158
IPv4 Broadcast Address on this Subnet	192.168.200.159

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Problem 2:

Given:	
Host IP Address:	10.101.99.228
Original Subnet Mask	255.0.0.0
New Subnet Mask:	255.255.128.0

Find:	
Number of Subnet Bits	9
Number of Subnets Created	512
Number of Host Bits per Subnet	15
Number of Hosts per Subnet	32,768
Network Address of this Subnet	10.101.0.0
IPv4 Address of First Host on this Subnet	10.101.0.1
IPv4 Address of Last Host on this Subnet	10.101.127.254
IPv4 Broadcast Address on this Subnet	10.101.127.255

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Problem 3

Given:	
Host IP Address:	172.22.32.12
Original Subnet Mask	255.255.0.0
New Subnet Mask:	255.255.224.0

Find:	
Number of Subnet Bits	3
Number of Subnets Created	8
Number of Host Bits per Subnet	13
Number of Hosts per Subnet	8192
Network Address of this Subnet	172.22.32.0
IPv4 Address of First Host on this Subnet	172.22.32.1
IPv4 Address of Last Host on this Subnet	172.22.63.254
IPv4 Broadcast Address on this Subnet	172.22.63.255

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Problem 4:

Given:	
Host IP Address:	128.107.0.55
Original Subnet Mask	255.255.0.0
New Subnet Mask:	255.255.255.0

Find:	
Number of Subnet Bits	8
Number of Subnets Created	256
Number of Host Bits per Subnet	8
Number of Hosts per Subnet	256
Network Address of this Subnet	128.107.0.0
IPv4 Address of First Host on this Subnet	128.107.0.1
IPv4 Address of Last Host on this Subnet	128.107.0.254
IPv4 Broadcast Address on this Subnet	128.107.0.255

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Problem 5:

Given:	
Host IP Address:	192.135.250.180
Original Subnet Mask	255.255.255.0
New Subnet Mask:	255.255.255.248

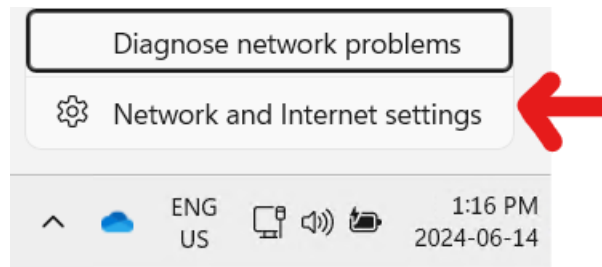
Find:	
Number of Subnet Bits	5
Number of Subnets Created	32
Number of Host Bits per Subnet	3
Number of Hosts per Subnet	8
Network Address of this Subnet	192.135.250.176
IPv4 Address of First Host on this Subnet	192.135.250.177
IPv4 Address of Last Host on this Subnet	192.135.250.182
IPv4 Broadcast Address on this Subnet	192.135.250.183

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B. Configuring the NIC

View network connection details

1. Right click the device summary in the task bar and select Network and Internet Settings.



This screen presents a brief overview of the network settings

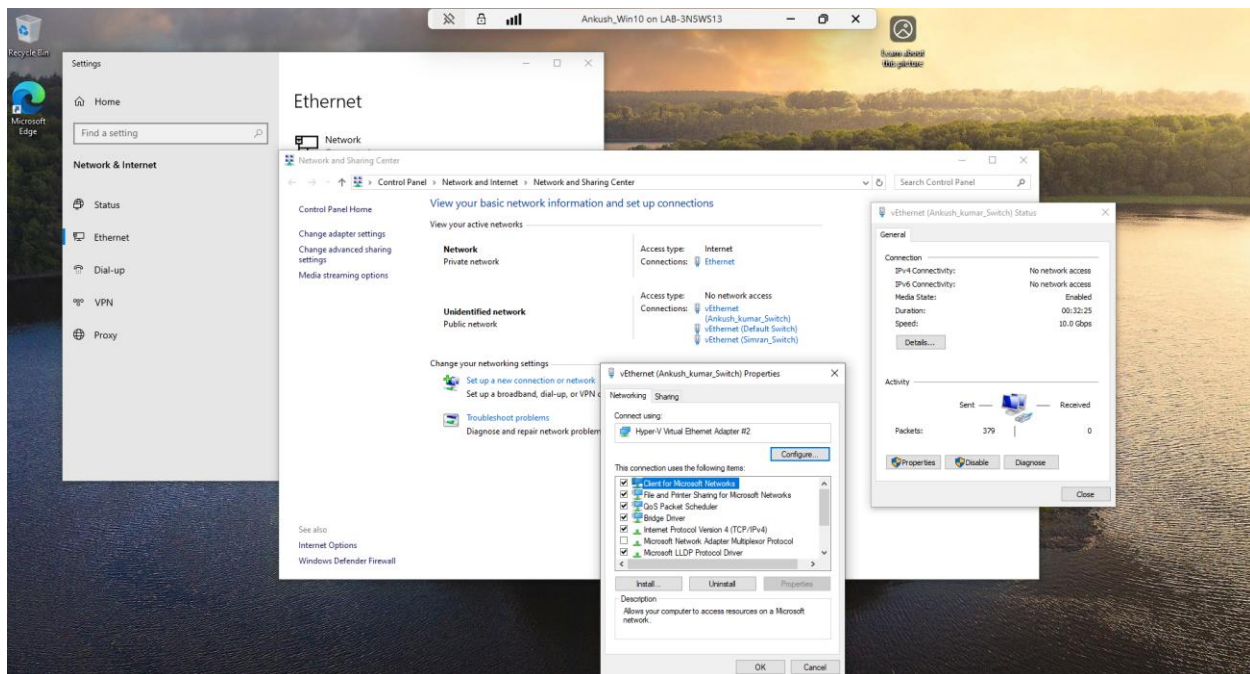
View network adapter properties

1. Select *Advanced network settings*.
2. Select the down arrow by the vEthernet connection. This is the Hyper-V Switch we made in Lab 4.



3. Select *Edit*.

<<SS01>> Take a screenshot of the vEthernet properties window.

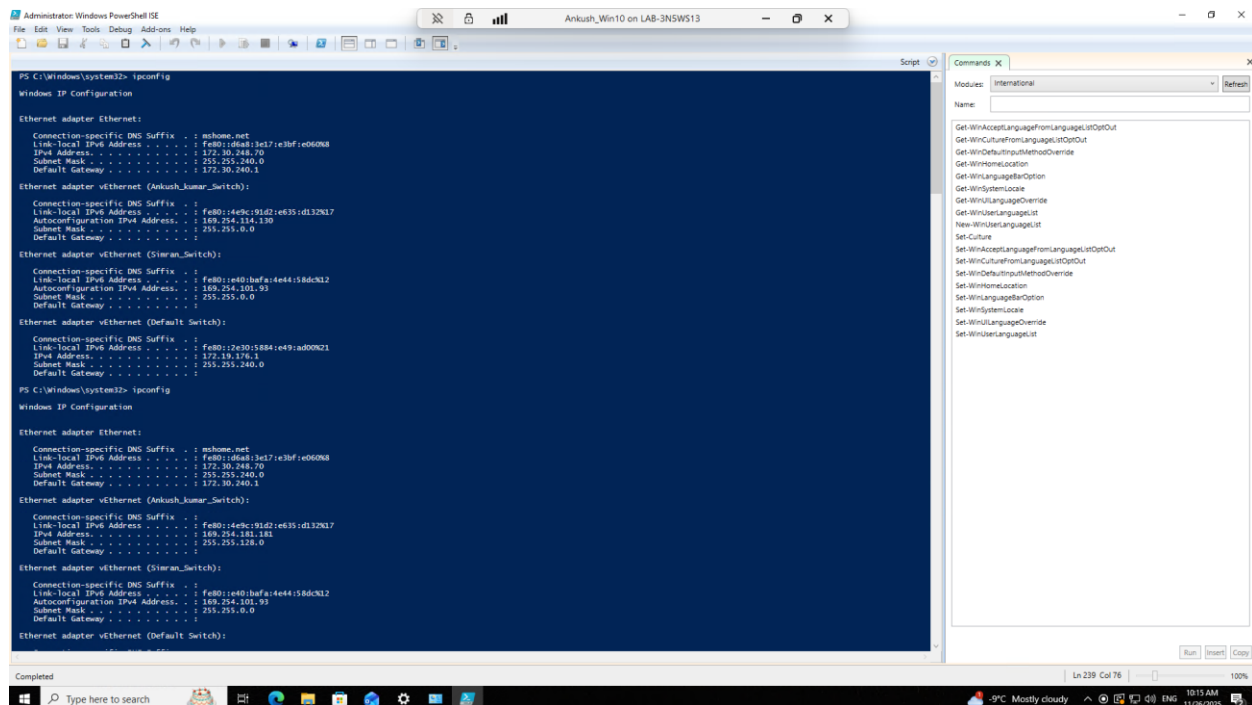


From this window we can adjust the configurations of the network adaptor.

Command line network information

1. Right click the windows logo and select Run.
2. Enter **PowerShell ISE** and click *Run*.
3. The Windows PowerShell ISE window will open. This utility allows you to use command line arguments. PowerShell ISE in Windows is an extremely powerful tool.
4. Type in the command **ipconfig**.
5. This return shows the same type of information in the connection details screen.

<<SS02>> Take a screenshot of the command line return for the vEthernet adapter.



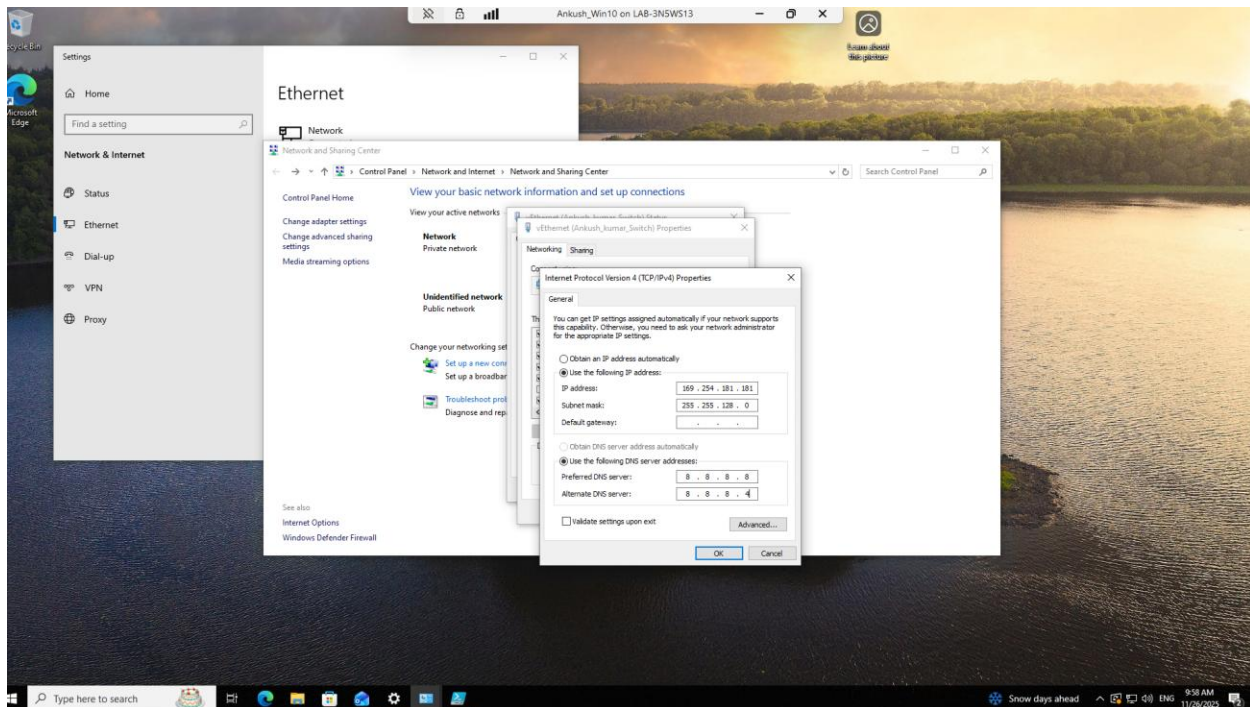
Change IP configuration

1. From the vEthernet properties window.
2. Select Internet Protocol Version 4 (TCP/IPv4).
3. Click Properties.
4. Click the **Use the following IP address** radio button.
5. Set IP address to **169.254.181.181**.
6. Set Subnet mask to **255.255.128.0**.

Change DNS (Domain Name Server)

1. If not already selected, click the Use the Following server addresses radio button.
2. Set the preferred DNS to **8.8.8.8**.
3. Set the Alternate DNS to **8.8.8.4**.
4. Click **OK**.

<<SS03>> Take a screenshot of properties for the vEthernet adapter.



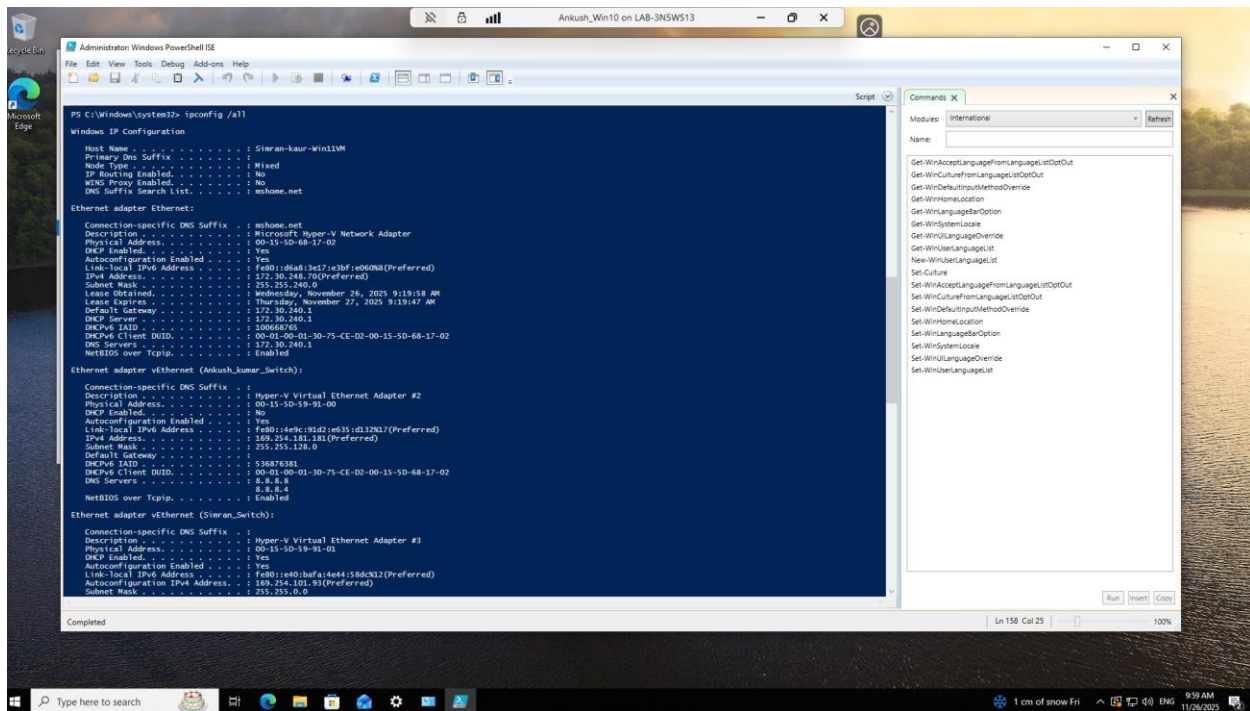
C. Other Network commands

The following commands are all to be done in PowerShell.

ipconfig

1. Enter the command ***ipconfig /help***.
2. This return shows the possible options that ipconfig can use, commands line /release and /renew can be used to reset the IP address by removing and requesting new ones from the DHCP server.
3. Enter the command ***ipconfig /all***.
4. This return gives additional information about the network adapters.

<<SS04>> Take a screenshot of the command line return for the vEthernet adapter.



ping

Ping can be used to find if a network address or name can resolve and be reachable.

1. Enter the command **ping 8.8.8.8**.
2. Enter the command **ping microsoft.com**.
3. Enter the command **ping FirstName_LastName_Ping.org**.
4. Note how the ping can resolve the IP address and a network name. Note that the FirstName_LastName_Ping.org should have errored out. If not, let your instructor know, as this is unexpected.

<<SS05>> Take a screenshot of all 3 ping requests.

```

Administrator: Windows PowerShell ISE
File Edit View Tools Debug Add-ons Help

Get-NetIPConfiguration
Link-local IPv6 Address . . . . . : fe80::e40:bafa:4e44:58dc%32(Preferred)
Autoconfiguration IPv4 Address . . . . . : 189.254.102.93(Preferred)
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : 189
DHCPv6 IAID . . . . . : 637519677
DHCPv6 Client DUID . . . . . : 00-01-00-00-30-75-CE-02-00-15-50-68-17-02
DNS Servers . . . . . : fe00:0:0:0::131
                        fe00:0:0:0::132
                        fe00:0:0:0::133
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter vEthernet (Default Switch):

Connection-specific DNS Suffix  : 
Description . . . . . : Hyper-V Virtual Ethernet Adapter
Physical Address . . . . . : 00-15-5D-A7-0F-3E
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::7e30:1888:e49:ad0%21(Preferred)
IPv4 Address. . . . . : 172.19.176.1(Preferred)
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . : 
DHCPv6 IAID . . . . . : 302127006
DHCPv6 Client DUID . . . . . : 00-01-00-00-30-75-CE-02-00-15-50-68-17-02
DNS Servers . . . . . : fe00:0:0:0::131
                        fe00:0:0:0::132
                        fe00:0:0:0::133
NetBIOS over Tcpip. . . . . : Enabled

PS C:\Windows\system32> ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=32ms TTL=114
Reply from 8.8.8.8: bytes=32 time=32ms TTL=114
Reply from 8.8.8.8: bytes=32 time=32ms TTL=114
Reply from 8.8.8.8: bytes=32 time=32ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 32ms, Maximum = 32ms, Average = 32ms

PS C:\Windows\system32> ping microsoft.com

Pinging microsoft.com [13.107.246.70] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 13.107.246.70:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

PS C:\Windows\system32> ping Ankush.kumar_Ping.org
Ping request could not find host Ankush.kumar_Ping.org. Please check the name and try again.

PS C:\Windows\system32>
Completed
Ln 185 Col 25 100%
  
```

tracert

Trace Route can be used to find the path a request will take through the network to get to its destination.

1. Enter the command ***tracert microsoft.com***.
2. This command may take a few minutes and may fail.

<<SS06>> Take a screenshot of the tracert return.

The screenshot shows a Windows PowerShell console window with the following commands and output:

```

PS C:\Windows\system32> ipconfig /all
ipconfig /all:
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . :
DHCPv6 IAID . . . . . : 352327005
DHCPv6 Client DUID. . . : 00-01-00-01-30-75-CE-02-00-15-5D-68-17-02
DNS Servers . . . . . : fec0:0:0:ffff::1%1
                       fec0:0:0:ffff::2%1
                       fec0:0:0:ffff::3%1
NetBIOS over Tcpip. . . . : Enabled

PS C:\Windows\system32> ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=12ms TTL=114
Reply from 8.8.8.8: bytes=32 time=12ms TTL=114
Reply from 8.8.8.8: bytes=32 time=12ms TTL=114
Reply from 8.8.8.8: bytes=32 time=12ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 32ms, Maximum = 32ms, Average = 32ms

PS C:\Windows\system32> ping microsoft.com
Pinging microsoft.com [13.107.246.70] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 13.107.246.70:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

PS C:\Windows\system32> ping Anush.kumar.Ping.org
Ping request could not find host Anush.kumar.Ping.org. Please check the name and try again.

PS C:\Windows\system32> traceroute microsoft.com
Tracing route to microsoft.com [13.107.246.70]
over a maximum of 30 hops:
  0  <1 ms <1 ms <1 ms LAB-3N5W513.nshome.net [172.30.240.1]
  1  * * * Request timed out.
  2  * * * Request timed out.
  3  * * * Request timed out.
  4  * * * Request timed out.
  5  * * * Request timed out.
  6  * * * Request timed out.
  7  * * * Request timed out.
  8  * * * Request timed out.
  9  * * * Request timed out.
 10  * * * Request timed out.
 11  * * * Request timed out.
 12  * * * Request timed out.
 13  * * * Request timed out.
 14  * * * Request timed out.

PS C:\Windows\system32> ]

```

The Command Prompt window on the right shows a list of Windows settings, including:

- Get-WinAcceptLanguageFromLanguageListOptOut
- Get-WinCultureFromLanguageListOptOut
- Get-WinDefaultInputMethodOverride
- Get-WinHomeLocation
- Get-WinLanguageBarOption
- Get-WinSystemLocale
- Get-WinUxLanguageOverride
- Get-WinUserLanguageList
- New-WinUserLanguageList
- Set-Culture
- Set-WinAcceptLanguageFromLanguageListOptOut
- Set-WinCultureFromLanguageListOptOut
- Set-WinDefaultInputMethodOverride
- Set-WinHomeLocation
- Set-WinLanguageBarOption
- Set-WinSystemLocale
- Set-WinUxLanguageOverride
- Set-WinUserLanguageList

route

Route can be used to view and edit the routing tables.

1. Enter the command **route print**.
2. This return shows the routing tables.

<<SS07>> Take a screenshot of the IPv4 Routing table, show as many addresses as possible.

Administration: Windows PowerShell

File Edit View Tools Debug Actions Help

Ankush_Win10 on LAB-3N5WS13

```

PS C:\Windows\system32> ipconfig /all

Interface List
0x{...} {...} {...} .....Microsoft Hyper-V Network Adapter
17...00 15 5d 59 80 00 .....Hyper-V Virtual Ethernet Adapter #2
12...00 15 5d 59 81 03 .....Hyper-V Virtual Ethernet Adapter #3
1...00 00 00 00 00 00 .....Software Loopback Interface 1
21...00 15 5d a7 0f 3e .....Hyper-V Virtual Ethernet Adapter

IPv4 Route Table

Active Routes:
Network Destination    Netmask          Gateway          Interface        Metric
0.0.0.0                0.0.0.0          172.30.248.1     172.30.248.70    15
127.0.0.0              255.0.0.0        On-link         127.0.0.1        331
127.0.0.1              255.0.0.0        On-link         127.0.0.1        331
127.255.255.255        255.255.255.255 On-link         127.0.0.1        331
169.254.0.0            255.255.0.0      On-link         169.254.101.93   271
169.254.101.93        255.255.255.255 On-link         169.254.101.93   271
169.254.181.181        255.255.255.255 On-link         169.254.181.181  271
169.254.181.181        255.255.255.255 On-link         169.254.181.181  271
169.254.255.255        255.255.255.255 On-link         169.254.101.93   271
172.15.176.0           255.255.240.0    On-link         172.15.176.1     5266
172.15.176.1           255.255.255.255 On-link         172.15.176.1     5266
172.30.248.0           255.255.240.0    On-link         172.30.248.70    271
172.30.248.70          255.255.255.255 On-link         172.30.248.70    271
172.30.255.255         255.255.255.255 On-link         172.30.248.70    271
224.0.0.0              240.0.0.0        On-link         227.0.0.1        331
224.0.0.0              240.0.0.0        On-link         169.254.181.181  271
224.0.0.0              240.0.0.0        On-link         169.254.101.93   271
224.0.0.0              240.0.0.0        On-link         172.30.248.70    271
255.255.255.255        255.255.255.255 On-link         127.0.0.1        331
255.255.255.255        255.255.255.255 On-link         169.254.181.181  271
255.255.255.255        255.255.255.255 On-link         169.254.101.93   271
255.255.255.255        255.255.255.255 On-link         172.30.248.70    271
255.255.255.255        255.255.255.255 On-link         172.15.176.1     5266

Persistent Routes:
None

IPv6 Route Table

Active Routes:
If Metric Network Destination Gateway
1 331 ::1 ::1/32 On-link
17 271 fe80::/64 On-link
12 271 fe80::/64 On-link
8 271 fe80::/64 On-link
21 5256 fe80::/64 On-link
12 271 fe80::e40:bafa:4e44:58dc/128 On-link
21 5256 fe80::2e30:5884:e49:a000/128 On-link
17 271 fe80::4e9c:91d2:e035:d132/128 On-link
8 271 fe80::d6a8:3e17:e3bf:e060/128 On-link
1 331 ::1/128 On-link
17 271 fe80::/64 On-link
12 271 fe80::/64 On-link
8 271 fe80::/64 On-link
21 5256 fe80::/64 On-link

Persistent Routes:
None
  
```

Commands

Modules: International Refresh

Name:

```

Get-WinAcceptLanguageFromLanguageListOptOut
Get-WinCultureFromLanguageListOptOut
Get-WinDefaultInputMethodOverride
Get-WinHomeLocation
Get-WinLanguageOption
Get-WinSystemLocale
Get-WinLanguageOverride
Get-WinUserLanguageList
New-WinUserLanguageList
Set-Culture
Set-WinAcceptLanguageFromLanguageListOptOut
Set-WinCultureFromLanguageListOptOut
Set-WinDefaultInputMethodOverride
Set-WinHomeLocation
Set-WinLanguageOption
Set-WinSystemLocale
Set-WinLanguageOverride
Set-WinUserLanguageList
  
```

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